HDOC Day-8

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SAP ID: 590033354
Subject: C programming
Date: 1-09-26

Question 1

Write a C program to find the material volume of a hollow sphere when the inner and outer radii are given.

Step 1: Understanding the Problem

The objective of this program is to calculate the volume of the material present in a hollow sphere. The user provides the outer and inner radii, and the program subtracts the inner sphere's volume from the outer sphere's volume.

Formula: $(4/3) \times \pi \times (R^3 - r^3)$

Step 2: Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	R=5,r=3	410.50	410.50	Pass
2	R=8,r=4	1876.58	1876.58	Pass
3	R=10,r=6	3284.00	3284.00	Pass

Step 3: Algorithm

1. Start

2. Read input values

R3. Apply the formula

4. Display the result

5. Stop

Step 4: Flowchart

Start

↓

Input

↓

Process

↓

Output

↓

Stop

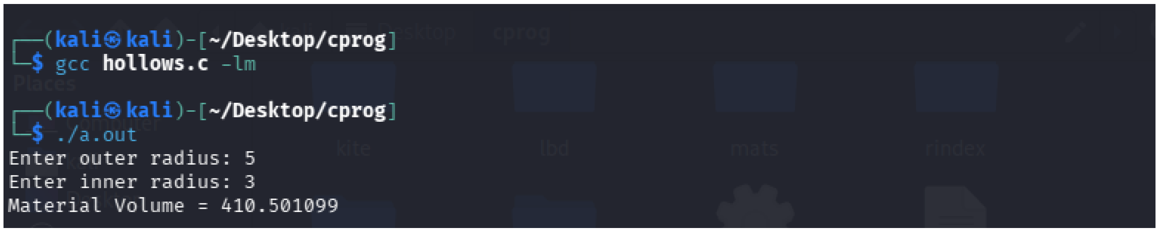
Step 5: C Program

```
#include <stdio.h>
#define PI 3.14159
int main(){
float R,r,v;
printf("Enter outer radius: ");
scanf("%f",&R);
printf("Enter inner radius: ");
scanf("%f",&r);
v=(4.0/3.0)*PI*(R*R*R-r*r*r);
printf("Material Volume = %f",v);
return 0;
}
```

Step 6: Compilation & Execution

Compile using GCC, execute the program, and compare the output with the prepared test cases.

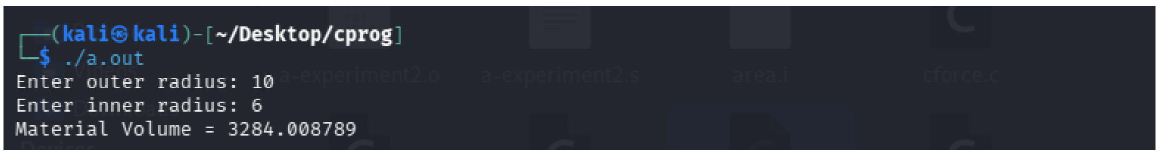
Step 7: Screenshots



```
┌──(kali㉿kali)-[~/Desktop/cprog]
└─$ gcc hollows.c -lm

┌──(kali㉿kali)-[~/Desktop/cprog]
└─$ ./a.out
Enter outer radius: 5
Enter inner radius: 3
Material Volume = 410.501099
```

```
┌──(kali㉿kali)-[~/Desktop/cprog]
└─$ ./a.out
Enter outer radius: 8
Enter inner radius: 4
Material Volume = 1876.576416
```



```
┌──(kali㉿kali)-[~/Desktop/cprog]
└─$ ./a.out
Enter outer radius: 10
Enter inner radius: 6
Material Volume = 3284.008789
```

Question 2

Write a C program to find the angle at the circumference when the central angle standing on the same arc is given.

Step 1: Understanding the Problem

This program determines the angle at the circumference using the central angle. According to the theorem, the angle at the circumference is always half of the central angle standing on the same arc.

Formula: Angle = Central Angle / 2

Step 2: Test Case Table

Test Case	Input	Expected Output	Actual Output	Result
1	60	30	30	Pass
2	90	45	45	Pass
3	120	60	60	Pass

Step 3: Algorithm

1. Start

2. Read input values

3. Apply the formula

4. Display the result

5. Stop

Step 4: Flowchart

Start

Input

Process

Output

Stop

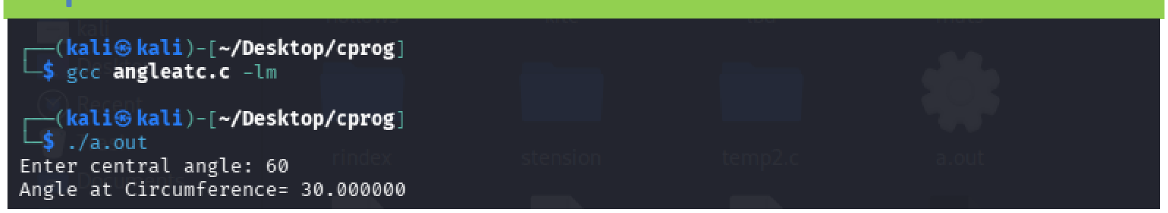
Step 5: C Program

```
#include <stdio.h>
int main(){
float c,a;
printf("Enter central angle: ");
scanf("%f",&c);
a=c/2;
printf("Angle at Circumference = %f",a);
return 0;
}
```

Step 6: Compilation & Execution

Compile using GCC, execute the program, and compare the output with the prepared test cases.

Step 7: Screenshots



```
┌──(kali㉿kali)-[~/Desktop/cprog]
└─$ ./a.out
Enter central angle: 90
Angle at Circumference= 45.000000
```

```
┌──(kali㉿kali)-[~/Desktop/cprog]
└─$ ./a.out
Enter central angle: 120
Angle at Circumference= 60.000000
```